Designing Adaptive LLM Scaffolding for Burmese-Speaking STEM Learners

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01 KEY FINDING

A working AI learning system is not necessarily a reliable tutor.

A bilingual STEM assistant combined terminology support, conceptual explanations and learner-requested assistance. Application controls worked in tested cases, but the **91 assessed outputs** included **18 pass · 71 partial · 2 fail**. Learning improvement was not measured.

Bilingual Terminology Support | **Conceptual Explanation** | **Adaptive Scaffolding**

02 OBSERVE

Language and conceptual barriers can occur together.

Burmese-speaking learners may struggle with specialised **English STEM terms** and unfamiliar concepts.

Literal translation may not preserve technical meaning or explain the idea.

In the **reviewed literature and systems**, language support, explanations and learner assistance were often separate.

Evidence base: 17 included studies · 5 reviewed systems

Research gap
How can these capabilities form **one structured, learner-responsive system**?

03 RESEARCH QUESTIONS

RQ1 — Terminology
How can specialised English STEM terminology be supported for Burmese-speaking learners?

RQ2 — Explanation
How can LLM-based support help learners understand STEM concepts beyond translation?

RQ3 — Adaptation
How can LLM-based scaffolding provide structured and adaptive support for Burmese-speaking STEM learners?

04 THEORISE

Task–Technology Fit (TTF)

Match language, explanation and interaction capabilities to learner tasks (Goodhue & Thompson, 1995).

Scaffolding Theory

Give structured, responsive help and consider when it can be reduced (van de Pol et al., 2010).

Learner Task → Suitable Technology → Requested, Structured Help

05 DESIGN | From design knowledge to an executable assistant

Conceptual
Core scaffolding concepts → task-aligned relationship model → context-aware adaptive framework.

System
Adaptive scaffolding system framework → logical architecture → Burmese STEM AI proof of concept.

Adaptive STEM Scaffolding Framework

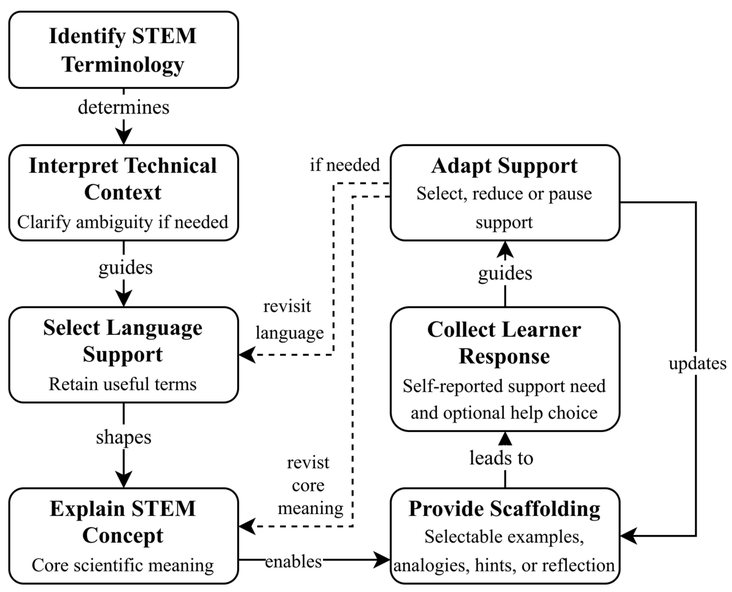


Figure 1. Context-Aware Adaptive STEM Scaffolding Framework.

Burmese STEM AI Prototype

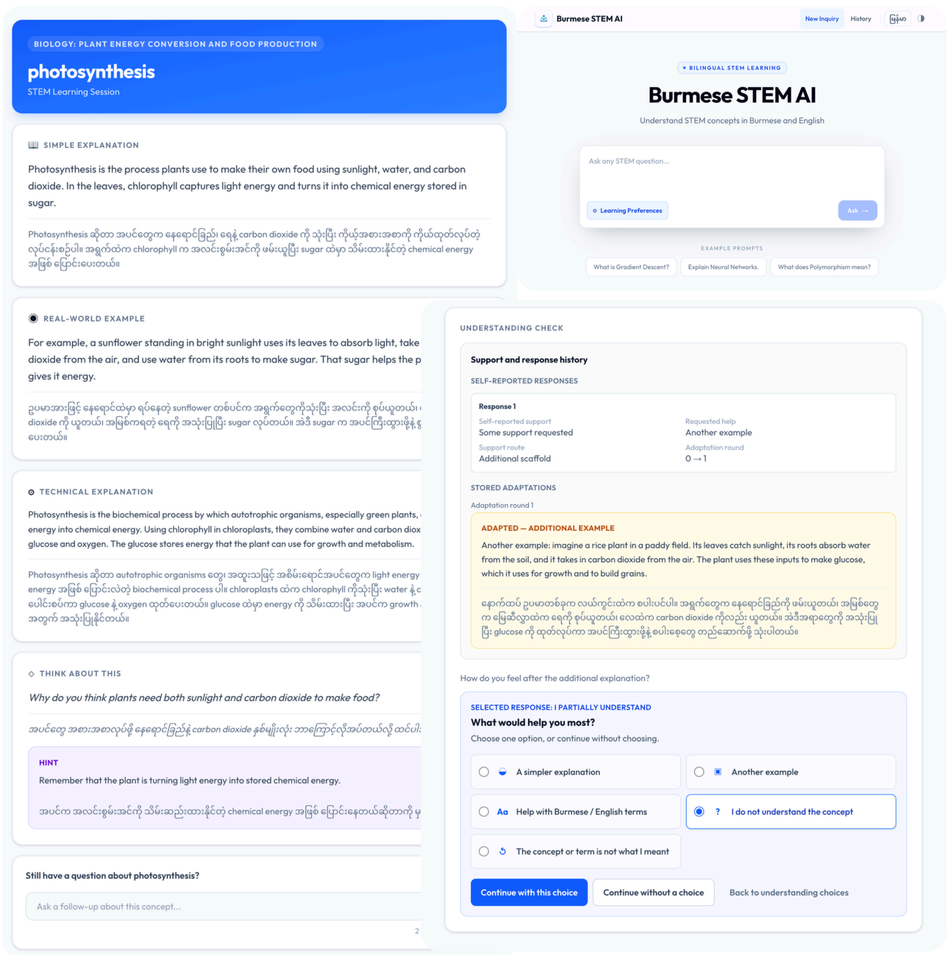


Figure 2. Burmese STEM AI prototype presents concept explanations in English and Burmese with structured learning support.

06 EVALUATE

Two complementary evaluation tracks

Conceptual artefact
- GenAI critique
- Literature comparison
- Informed argument
- Scenario

System artefact
- Functionality/usability check
- Static/dynamic analysis
- Bounds analysis
- Simulation
- Black-box/white-box testing
- Literature comparison
- Scenario

Distinguish:
Software behaviour ≠ Content quality ≠ Learner outcomes.

Results

Content assessment — 91 delivered outputs

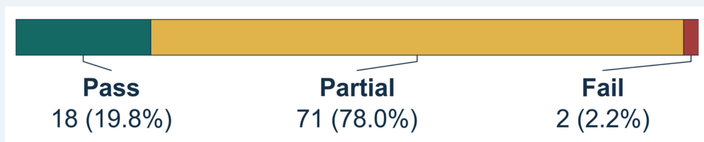


Figure 3. Content-quality ratings for 91 generated responses. Ratings were assigned by the author without an independent second reviewer.

Software and usability: All **518** controlled tests of program logic and **17 tests of system limits** passed. Usability inspection rated **6 of 9** criteria pass and **3** partial.

Weaknesses: Scientific and Burmese-language errors remained in saved explanations. Two simultaneous model requests produced **one saved revision**.

Interpretation: Tests supported specific software controls. **Consistent content accuracy and improved learning were not demonstrated.**

07 REFLECT & LEARN

Three lessons from the tested system

- **Valid format ≠ correct meaning**. Review scientific accuracy and Burmese meaning.
- **More help ≠ better help**. More support did not correct earlier errors. User requests do not mean mastery.
- **Saved limits ≠ spending limits**. Two saved revisions did not prevent duplicate provider calls.

08 GENERALISE

Five proposed design principles

DP1: Clarify technical meaning before choosing language support.
DP2: Use learner requests to guide help, not prove understanding.
DP3: Check scientific accuracy, bilingual consistency and response structure separately.
DP4: Record what changed, what remains unclear, what help was requested, and whether the session is complete.
DP5: Distinguish limits on saved revisions, model calls, time and cost.

09 CONTRIBUTE

Provisional improvement in design knowledge

- Combines **STEM terminology support, concept explanations and learner-requested help** in one evaluated system.
- Proposes **five conditional design principles** for clear support decisions, content checks, session records and interaction limits.
- Builds on **TTF and Scaffolding Theory**. New theory, unique novelty and better learning outcomes were not demonstrated.

Limitations

- No recruited learner experiment or independent bilingual subject review.
- Learning, retention, fit of support levels and transfer to other settings remain untested.

Scan for Live Demo

https://burmesestemai.serveirc.com/

Next

Independently verify explanations → Compare learner outcomes with a baseline → Evaluate support/cost limits → Test transfer to another language and subject

References

Goodhue, D. L., & Thompson, R. L. (1995). Task-technology fit and individual performance. *MIS Quarterly, 19*(2), 213–236. https://doi.org/10.2307/249689

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